COMPOSITE MATERIAL DATASHEETS CFRP AS4/APC2-UD

MATERIAL IDENTIFICATION

Short informal name	AS4/APC2-UD
Commercial name	Cytec/Solvay AS4/APC-2
Specifications	(*) This material is a not qualified according Airbus standards and specifications. The use of the design data presented in this report is restricted to R&T projects, and to preliminary design phases of airframes development. Design values not validated.
Description	Carbon fiber tape standard modulus (AS4) in thermoplastic resin. Produced by an In-Situ-Consolidation process. Service temperature: -55 to >70ºC (TBC).
Version	0.3

REFERENCES

- TAE-T-NT-150133 Is. D “Thermoplastic AS4/APC-2 with ISC mechanical properties.”
- TEA-CS-NT-170067 Is. B “AS4/APC-2 (PEEK thermoplastic) Compression After Impact CAI. TEST RESULTS ANALYSIS”
- TEA-CS-NT-170070 Is. B “ANALYSIS OF CYTEC THERMOPLASTIC UD AS4/APC2 BOLTED JOINT COUPONS TEST RESULTS AND ALLOWABLE VALUES DERIVATION”
- TEA-CS-NT-180111 Is. A “RESIDUAL STRENGTH AFTER IMPACT OF CFRP CYTEC/SOLVAY AS4/APC-2 (ISC) UD-TAPE LAMINATES”
- TE-T-MM-230027 Is. A “MATERIAL DATA INPUTS FOR SIMULATION OF COMPOSITE CRIPPLING TEST AT PRINCIPIA”
- “APC-2-PEEK Thermoplastic Polymer”, Cytec technical data-sheet

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.135	mm
Density	1320	kg/m³
Tg-onset (wet)		ºC
Max. Operational Limit (MOL)		ºC

Coefficients of Thermal Expansion (1/ºC)	Range	-55..20ºC	20..70ºC	70..120ºC
	α_1			
	α_2			
	α_3			

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
Values	120000	8800	3800	0.35	0.88

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13} -	ν_{23} -
Values					

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1, E_2 and G_{12} in buckling stability analyses

PLAIN STRENGTHS

Property	Mean Values (MPa)				B-basis	*E-KDF*			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{1t}	2080				0.89	1.0		0.60	1851	1851		1110
F_{1c}	1050				0.83	0.95		0.90	871	827		784
F_{2t}	49.3				0.83	1.0		0.60	40.9	40.9		24.6
F_{2c}	165				0.83	0.95		0.75	137	130		102
F_{12}	148				0.83	0.78(*)		1.0	122	(*)		122

(*) Test result not following the expected trend

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (με)				B-basis	*E-KDF*			B-basis Values (με)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
ε_{tall}	17300				0.89	1.0		0.80	15400	15400		12300
ε_{call}	8750				0.83	0.95		0.90	7260	6900		6500

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis	*E-KDF*			B-basis Values			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{13} (MPa)	75				0.83	1.0		0.60	62	62		37
F_{3t} (MPa)												
G_{IC} (N/mm)	1.9											
G_{IIC} (N/mm)	2.0											

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis	*E-KDF*			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
OHT_{QI}	386				0.89	1.0		0.90	343	343		309
OHC_{QI}	291				0.83			0.60	241			145
FHT_{QI}	394				0.89	1.0		0.90	350	350		315
FHC_{QI}	420				0.83			0.78	348			272
$F_{BR\text{-}QI}$	819				0.83	0.94		0.90	680	639		612
F_{PO}												

(*) Per similarity with PT tests results, merging data of *UniTow* and *MultiTow* processes.

COMPOSITE DAMAGE TOLERANCE

Design value for tension residual strength after BVID impact.

Property	B-basis Value for all conditions
TAI (με)	4700

CAI model: $CAI = K_B \cdot E\text{-}KDF \cdot [\, CAI_{\infty\text{-}QI} - C \cdot (E_x/E_{qi} - 1) + A / e^{B \cdot (E/t^{\wedge}2)} \,]$

Property	Mean Value	B-basis	E-KDF		
	RT/Dry	K_B	70ºC/W	90ºC/W	120ºC/W
$CAI_{\infty\text{-}QI}$ (με)	4400	0.83	1.0		
A (με)	7200				
B (mm²/J)	0.83				
C (με)	2700				

Use as cut-off for the CAI strength the allowable strains shown above for intact material (ε_c).